Final Project Presentation

Investigating AI Adoption in Construction Companies

Literature review + industry interviews + practical implications

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Why this topic matters

AI is moving from hype to practical adoption in AEC, but implementation is uneven.

- AEC firms want productivity gains, better decision support, and less repetitive administrative work.
- Adoption is still slowed by fragmented data, variable workflows, privacy concerns, and governance gaps.
- This project combines academic frameworks with interview evidence to identify where AI is already useful and what limits broader rollout.

Practical focus | Interview-based | AEC-specific

Project questions

1. Where is AI delivering meaningful value today?

2. What barriers are slowing adoption on real projects?

3. What should firms do first for practical ROI?

Research approach

Qualitative exploratory study using literature plus two semi-structured interviews.

1. Literature review

- AI adoption frameworks: DOI, TAM, UTAUT/UTAUT2, TOE
- Construction-specific studies on opportunities, barriers, and skills
- Focus on practical implementation, not just theory

2. Interview evidence

- Phone interview: Dane Construction leader
- Email interview: Australian AEC expert-practice perspective
- Questions covered tools, barriers, training, outcomes, and future strategy

3. Thematic analysis

- Drivers
- Barriers
- Organizational support
- Impact
- Future outlook

Note: the second interview strengthens sector insight but is best treated as a supplementary expert perspective rather than a second named company case.

Literature foundation: what explains adoption?

Four frameworks explain why AI spreads unevenly across firms and workflows.

DOI

Diffusion of Innovations

Adoption spreads through innovators and early adopters before broader uptake.

TAM

Technology Acceptance Model

Use depends on perceived usefulness and ease of use.

UTAUT 2

Unified Theory of Acceptance

Adds social influence, facilitating conditions, habit, and value.

TOE

Technology–Organization–Environment

Adoption depends on technical fit, internal readiness, and external pressures.

Main takeaway: AEC adoption is not only about the AI tool itself. It also depends on workflow fit, skills, governance, and organizational readiness.

Where AI is delivering value today

The strongest current value is in lower-risk, information-heavy workflows.

High-value use cases

- Tender summarization, spec review, and contract intelligence
- RFI drafting, meeting-minute extraction, and correspondence support
- Bid analysis and project knowledge search across drawings, emails, and reports
- Safety monitoring, PPE compliance, defect tagging, and evidence capture
- Image-based verification and searchable site records

GenAI | Computer vision | Knowledge search | Defect review

Interview snapshot

The two interviews show similar themes from different angles.

Dane Construction

Regional U.S. contractor | leadership perspective

- Current tools: home-built estimating system, ChatGPT, Copilot
- Main value: drafting, polishing, meeting notes, bid-data analysis
- Leadership is supportive but cautious about confidentiality and ownership
- Employee usage is still lower than expected
- AI seen as a time-saving augmentation tool

Australian AEC expert

Expert-practice perspective | sector-wide view

- Strongest value: document intelligence + computer vision workflows
- Highlights safety monitoring, defect tagging, and knowledge search
- Stresses messy real-world data vs clean academic assumptions
- Sees strong future potential in vision-language models
- Recommends starting with practical, measurable ROI use cases

What is slowing adoption?

The main barriers are organizational and data-related, not just technical.

Data fragmentation	Project information is spread across contractors, consultants, subs, and software systems.
Governance gaps	Firms still need clear rules for confidentiality, copyright, liability, and review.
Workflow variation	A tool that works on one project may not transfer easily to another job.
Trust and culture	Users are cautious when outputs affect safety, contracts, or compliance.

Shared message from both interviews: use AI, but verify it.

Overhype vs practical value

The interviews separate marketing claims from current working use cases.

Overhyped right now

- Fully autonomous jobsites
- “AI project manager” claims
- Automatic 4D BIM progress tracking without human validation
- Generic chatbots trained on poorly structured project data

Actually working

- Document review and tender summarization
- Meeting summarization and drafting support
- Drawing/specification search and knowledge retrieval
- Photo-based safety and quality support

Recommendations for immediate ROI

Start small, measurable, and governed.

1. Start with document intelligence, meeting summaries, knowledge search, and evidence capture.
2. Set governance rules early for confidentiality, copyright, acceptable use, and human review.
3. Train for AI literacy and workflow redesign, not just tool access.
4. Use AI as augmentation for contracts, safety, quality, and compliance—not as a replacement for judgment.
5. Improve data readiness gradually so better models have a stronger foundation.

Key takeaways

AI already has value in AEC, but success depends on readiness more than hype.

- The strongest current value is in document-heavy and evidence-heavy workflows.
- The biggest barriers are fragmented data, governance uncertainty, workflow fit, and trust.
- Firms should start with practical use cases and scale only with human verification and clear rules.
- AI in AEC is best understood today as augmentation—not autonomous project delivery.

Thank you